STATUS Open to Senior SWE & Staff / Technical PM roles

TEHRAN, IRAN · REMOTE

Alireza Jafartash

Senior Software Engineer · Systems Architecture & Financial Ledgers

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00 PROFILE

Computer Engineering graduate student at University of Tehran specializing in algorithms, computations, and distributed systems. 8+ years of engineering craftsmanship designing high-throughput transaction backends, multi-tenant B2B2C platforms, and financial ledgers. Proven track record architecting mission-critical systems in production across Go, Java (Spring Boot), TypeScript, and PostgreSQL. Combines deep algorithmic problem-solving with pragmatic product ownership.

01 CORE ARSENAL

LANGUAGES	Go · TypeScript / Node · Java 21 (Spring Boot 3) · Python · SQL · Kotlin · C++
SYSTEMS & LEDGERS	High-Throughput Ledgers · Multi-Tenant B2B2C · Concurrency & Locks · Double-Entry Accounting · Ka
DATABASES & CACHE	PostgreSQL · Redis (Geospatial & Time-Series) · Liquibase · Drizzle ORM · Query Optimization
CLOUD, INFRA & IAC	Google Cloud Platform (GCP) · Docker · Kubernetes · Terraform · Cloudflare Workers · Bitbucket /
ARCHITECTURE & AI	Microservices · Clean Architecture · Agentic AI CLI Multiplexing · REST & GraphQL APIs · Typst

02 NOTABLE TOOLS

Promux ↗

Go · CLI · AI Systems

Autonomous multiplexer for agentic AI CLIs with zero-downtime account rotation.

Architected a proxy multiplexer that intercepts agentic CLI calls. Automatically detects HTTP 429/500 quota exhaustion, triggers account rotation with TTL cooling, and preserves running agent session context without interrupting long code generation loops.

Manova / Orbit Platform ↗

TypeScript · Go · PostgreSQL · Edge

Standardized foundation platform for AI-native and client microservices.

Production micro-platform providing unified RBAC authentication, token-bucket rate-limiting, double-entry payment verification, and multi-tenant DB partitions across 5+ client applications.

Dorost ↗

Typst · Open Source · Design System

Open-source bilingual document composition package on official Typst registry.

Authored and published an open-source Typst package providing zero-dependency RTL/LTR mixed-direction layout, numeral conversion, and design system components with automated CI test suites.

Distributed UUID Generation System ↗

Go · RabbitMQ · Hetzner Cloud

High-throughput distributed 32-byte UUID generation platform.

Engineered a distributed ID generation system capable of producing collision-free 32-byte UUIDs under massive concurrent request volumes, with automated node coordination.

03 EDUCATION

Master of Computer Engineering — Algorithms and Computation — 2024 — PRESENT
University of Tehran
Head Teaching Assistant for Advanced Programming · Research in distributed algorithms

Bachelor of Computer Engineering — 2018 — 2024
Babol Noshirvani University of Technology
Final Project: Distributed UUID Generation System

04 HONORS

- 2nd Place — Golang (Bale) Software Engineering Selection — 2023
- Top 20 — DevOps (Digikala) Boot Camp Selection — 2022
- Top 30 — CyberSecurity (Quera) Competition — 2023
- Top 30 — Software Engineering (Digikala) Bootcamp — 2022
- Top 20 — DiffCode (Quera) Algorithmic Contest — 2019
- Top 300 — Codeforces Round 611 (Div. 3) — 2019

05 EXPERIENCE

Lead Systems Architect (Freelance)
2025 — PRESENT

Independent / Enterprise Clients — Tehran

Architecting mission-critical platforms and automations for enterprise businesses.

- Architected Goldstein: multi-tenant SaaS gold trading platform with double-entry wallet accounting, real-time OTC desk transfers, and automated 750-fineness normalization.
- Engineered multi-branch Restaurant Inventory Management System: real-time stock ledger, automated supplier purchase orders, and recipe yield tracking.

TECH: Next.js 16 · React 19 · NestJS · Bun · PostgreSQL · Tailwind 4 · Redis

Founder & Lead Systems Engineer
2024 — PRESENT

Manova — Remote

AI workflows, shared platform engineering, and client infrastructure.

- Architected Orbit: shared micro-platform powering all client products with unified RBAC auth, token-bucket rate limiting, billing journals, and multi-tenant provisioning.
- Engineered Promux: zero-downtime agentic CLI multiplexer with automated account rotation on 429/500 errors and state preservation.

TECH: Go · TypeScript · Bun · PostgreSQL · Docker · Cloudflare Workers · Typst

Head Teaching Assistant — Advanced Programming
2024 — PRESENT

University of Tehran — Dept. of Science — Tehran

Leading teaching staff and curriculum infrastructure for competitive CS engineering course.

- Manage and coordinate a team of 15+ TAs responsible for problem design, grading, and curriculum delivery for 200+ students per semester.
- Designed and maintain the automated containerized judge system and assignment evaluation toolchain used across all terms.

TECH: Java · Automated Judging CI · Docker · Concurrency · OOP Design Patterns

Software Engineer
AUG 2024 — 2025

LvtGames — Turkiye (Remote)

High-concurrency gaming and entertainment platform.

- Engineered high-throughput transaction processing and multi-balance wallet management on a multi-tenant B2B2C entertainment platform.
- Operated GCP infrastructure via Terraform IaC, built Bitbucket caching CI/CD pipelines, and led cross-functional feature launches as Product Manager.

TECH: Java 21 · Spring Boot 3 · Go 1.24 · PostgreSQL · Redis · GCP · Terraform · Liquibase

Software Engineer
MAY 2025 — JUL 2025

SwapWallet — Tehran

Seamless crypto trading exchange and wallet platform.

- Optimized high-throughput crypto exchange engine for low latency under peak concurrent loads; resolved complex concurrency race conditions to guarantee ledger integrity.
- Designed and maintained key operational utility Telegram bot streamlining administrative and member workflows.

TECH: Apache Kafka · Kotlin · Kubernetes · Spring Framework · PostgreSQL

Software Engineer
FEB 2024 — OCT 2024

Ultratrader — Tehran

Smart trading journal & portfolio manager for financial markets.

- Designed a high-throughput data preprocessing pipeline for streaming and normalizing high-volume financial market data across multiple broker APIs.
- Developed high-performance REST/GraphQL APIs, implemented resilient broker proxy gateways, and optimized heavy analytical database queries.

TECH: NestJS · PostgreSQL · Redis · Docker · GraphQL

Software Engineer
MAY 2022 — DEC 2023

Ravindex — Tehran

Financial analysis platform for Tehran Stock Exchange.

- Processed high volumes of precise stock market tick data with optimized analytical ingestion pipelines and replicated PostgreSQL databases.
- Developed microservices with RabbitMQ message queues and authored 20+ automated Lambda functions for data fetching with Prometheus/Grafana telemetry.

TECH: Django · PostgreSQL · RabbitMQ · Docker · Prometheus · Grafana

Systems & R&D Engineer
2016 — 2022

Earlier Engineering: Plotset & Cubecraft — California, US & London, UK (Remote)

AI visual tools R&D and high-scale multiplayer game network engineering.

- Plotset (US): Researched AI-driven visualization solutions; architected unified framework integrating multiple charting engines.
- Cubecraft (UK): Built minigame engine plugins for high-scale Minecraft network with 300+ event listeners and a 50+ state finite state machine.

TECH: Node.js · D3.js · Java · SpigotMC API · MongoDB

06 CREDENTIALS

Software Engineering Bootcamp
Digikala · Sep 2022
Intensive training in advanced backend development, API design, database architecture, design patterns, and cross-functional communication.

Machine Learning Specialization
DeepLearning.AI · Mar 2023
Comprehensive study of foundational machine learning, neural networks, loss optimization, and engineering practices for building ML systems.

Cybersecurity Advanced Workshop
Quera & National Banking Specialists · May 2023
Practical security training with national banking specialists covering web security, reverse engineering, and applied cryptography.